

Prime Factors, HCF & LCM

WorksheetHub AI
Mathematics · Grade 7

Unit 1 · Number

Medium

Factors, multiples, HCF and LCM

Name: _____

Date: _____

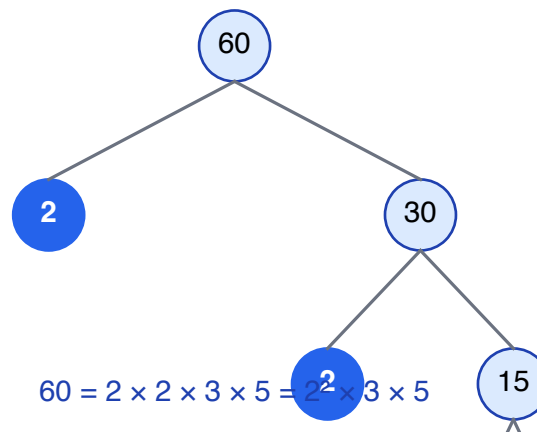
Score: / 13

Remember

Factors divide exactly into a number. **Multiples** are in its times table.

HCF (Highest Common Factor) = the biggest number that divides into both.

LCM (Lowest Common Multiple) = the smallest number in both times tables.



A factor tree: keep splitting until every branch ends in a prime (dark blue)

Worked example: HCF and LCM of 12 and 18

Factors of 12: 1, 2, 3, 4, 6, 12 | Factors of 18: 1, 2, 3, 6, 9, 18 → common: 1, 2, 3, 6 → **HCF = 6**

Multiples of 12: 12, 24, 36... | Multiples of 18: 18, 36... → **LCM = 36**

A. Write as a product of prime factors

1 Write **36** as a product of primes.

Answer:

2 Write **48** as a product of primes.

Answer:

3 Write **90** as a product of primes.

Answer:

4 Write **100** as a product of primes.

Answer:

B. HCF and LCM

- 5 Find the HCF and LCM of **12** and **18**.

Answer:

- 6 Find the HCF and LCM of **24** and **36**.

Answer:

- 7 Find the HCF and LCM of **15** and **20**.

Answer:

- 8 Find the HCF and LCM of **16** and **40**.

Answer:

C. Word problems

- 9 Bus A leaves the station every 12 minutes. Bus B leaves every 18 minutes. They both leave at 8:00. When do they next leave together?

Answer:

- 10 A teacher has 24 pencils and 36 erasers. She makes identical packs using all of them. What is the greatest number of packs she can make? How many of each item are in a pack?

Answer:

- 11 Two lights flash every 8 s and every 20 s. They flash together now. After how many seconds will they flash together again?

Answer:

★ Challenge

- 12 The HCF of two numbers is 5 and their LCM is 60. One number is 15. What is the other?

Answer:

- 13 Find the smallest number that is divisible by 2, 3, 4, 5 and 6.

Answer: